

EmoPus: Providing Emotional and Tactile Comfort with a AI Desk Companion Octopus

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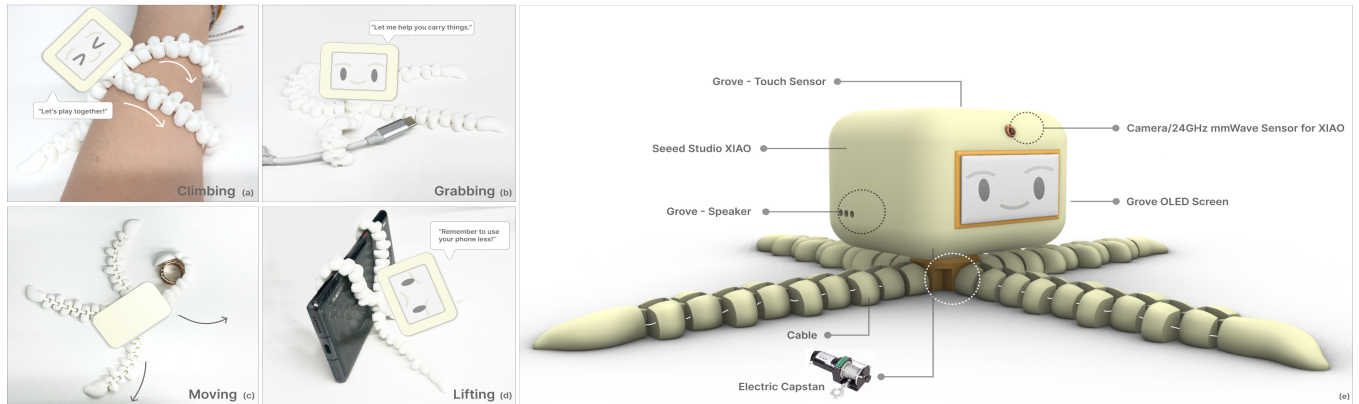


Figure 1: Overview of EmoPus

ABSTRACT

In the desktop scene, people often work and study for a long time, which can lead to mental stress and depression, therefore bringing the need for companion robots. Existing products offer voice dialogue and motion changes but lack direct user interactions. Consequently, we propose EmoPus, an AI companion robot resembling an octopus, designed for multi-dimensional interactions. Users can talk to EmoPus by voice and command it to complete simple tasks like grabbing objects. It also actively interacts based on the user's mental state, soothing the user's arm with its tentacles and offering a comforting tactile perception. User interaction would also change the shape of the EmoPus itself and provide real-time responses. In general, EmoPus interacts with users based on their psychological state, bringing fun, mental relief, and Emotional Companion while reducing the psychological burden after long work or study hours.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*;

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KEYWORDS

Emotional Companion, AI Robot, Tactile Comfort

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1 INTRODUCTION

In today's fast-paced work environments, maintaining a balance between productivity and mental well-being is crucial. Desk-bound settings often lead to stress, physical discomfort, and a lack of physical activity, which can negatively impact both mental health and productivity.

Currently, available AI desk companions are limited to interactions through voice, digital screens [2], and projection [1]. However, they do not engage in more in-depth interactions with users. Through tactile interaction, users can feel "soothed" and experience a more intimate touch interaction [6].

Various approaches have been explored in current research on driving 3D-printable mechanical structures. For instance, inter-meshing motion transfer mechanisms are used to control the motion of 3D models [5], but they are affected by self-weight and friction between kinematic components. Other methods involve pneumatic

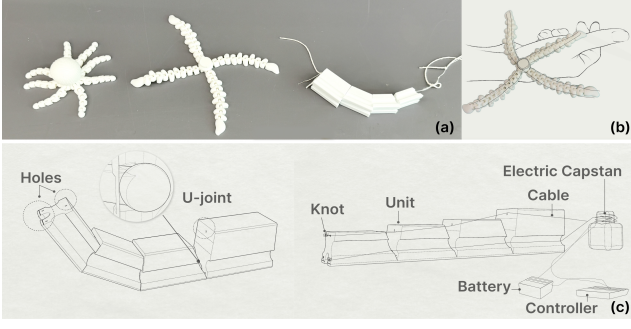


Figure 2: (a) Prototype construction; (b) Interaction forms; (c) Actuation Mechanisms of Tentacles

systems [7], utilizing negative pressure and high-frequency actuation with specific deformation structures [3]. However, these methods often suffer from low payload capacity and require multiple driving components (motors). For our project, we aim to implement a cable-driven approach [10]. This method offers advantages such as reducing the overall size of the structure while meeting our specific interaction requirements for touching, grasping, and crawling interactions with humans.

In this paper, we propose EmoPus, a desk-based companion robot. Inspired by biological behavior, the octopus-like design aims to mimic natural interactions, providing a deeper emotional connection. Moreover, we present prototypes of its tentacle modules. The potential contributions of this robot include: (1) Innovating interaction methods by providing tactile comfort and emotional support through a cable-driven 3D structure. (2) Assisting with simple tasks like grabbing objects on the desk through its flexible modular structure. (3) Fine-tuning large language models (LLMs) to provide emotional companionship.

2 DESIGN IMPLEMENTATION

During our prototyping stage, we experimented with various actuation methods and appearances for the octopus tentacles, as illustrated in appendix Figure A.2. We 3D-printed a series of cable-driven soft robots capable of actively controlling the curling direction of the tentacles, as shown in Fig. 2. The modules and lengths of the tentacles can be customized.

2.1 Voice Interaction

Voice interaction is crucial in companionship, as it responds to users' questions and provides appropriate emotional responses. It projects empathy through voice, triggering user empathy and promoting emotional connection [8]. Therefore, we designed EmoPus to incorporate emotion and content recognition capabilities. EmoPus listens to and responds to voice commands using Grove Vision AI Module V2 and PDM microphones, making it a highly interactive desktop companion. Users can call EmoPus or start a conversation via LLM by using XIAO ESP32C3 microcontroller, reducing isolation during desk-bound work. Speech emotion recognition results are applied to the robot's dialogue and actions, with its tentacles changing shape according to the user's emotions. As shown in Appendix Figure A.1, users can wake up EmoPus by calling it by

its name. When a user expresses joy, EmoPus' tentacles will be placed on the user's hand, enhancing the user's emotional experience. Conversely, a sad tone triggers a more subdued response, with tentacles opening gently, providing comfort and empathy through its actions.

2.2 Physical Interaction

Physical contact and interaction provide a real emotional connection. A servo motor controls all the physical interactions, allowing the tentacles to hinge. Tactile feedback helps users feel noticed and understood, enhancing their emotional security and fostering intimate relationships [4]. EmoPus features Grove touch sensors that detect when a user taps or presses it, as illustrated in Appendix Figure A.1. A double tap activates the "Companion Mode," where EmoPus engages in activities designed to provide companionship and reduce feelings of loneliness. A long press, on the other hand, activates "Focus Mode," helping the user concentrate on tasks by minimizing distractions. These interactions enable users to engage with EmoPus physically, fostering a sense of connection and presence.

2.3 Emotional Expression

EmoPus displays facial expressions that react to users' voices and facial emotions to enhance interaction. Referencing Appendix Figure A.1, EmoPus lights up with a joyful expression when the user actively interacts with it, thereby reinforcing positive engagement. If ignored, EmoPus displays a sad face on its LCD screen, subtly encouraging the user to re-engage. This feature aims to create an emotional bond between users and EmoPus, transcending its mechanical function. EmoPus uses advanced sensors to detect the user's presence and emotional state. The 24GHz mmWave sensor and Grove Vision AI module work together to recognize static human presence and visual cues related to emotions. This emotional feedback loop helps to create a dynamic and responsive interaction, enhancing the emotional bond between the user and EmoPus.

2.4 Task Assistance

A well-designed companion robot should provide both mental relief and practical functionality. In a desk environment, users often perform tedious tasks that can be exhausting[9]. EmoPus is not just a companion but also an assistant, capable of responding to speech commands related to various tasks. According to Appendix Figure A.1, EmoPus' tentacles can grasp and move objects by using a 24GHz mmWave sensor and camera to detect objects, thereby assisting the user with physical tasks at their desk. This enhances convenience and reduces the need to get up frequently. By extending its utility beyond emotional support, EmoPus becomes a practical tool for productivity, making it an invaluable addition to the workspace.

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